

PSEUDOCODES

1. Surface Blasting Design Calculator

INPUT:

Blasting Diameter, Average In Hole

Density, Spacing To Burden Ratio, Bench Height

Proximity and Rock Type .

- Proximity TO UPPERCASE

- Rock Type TO UPPERCASE

- $\text{Burden} = \text{Mass of Explosive} (\text{BenchHeight} - \text{StemmingLength}) / \text{Spacing} - \text{to} - \text{BurdenRatio} \times \text{TechnicalPowderFactor}$

$\text{Mass of Explosive} = \text{AverageIn} - \text{HoleDensity} \times (\text{BlastingDiameter})^2 / 1273 (2)$

$\text{Spacing} = \text{Spacing} - \text{to} - \text{BurdenRatio} \times \text{Burden} (3)$

- Proximity = "CLOSE TO COMMUNITY"

- $\text{Stemming Length} = 30 \times (\text{Blasting Diameter})$

- IF Proximity = "NOT"

- $\text{Stemming Length} = 20 \times (\text{Blasting Diameter})$

- IF Subdrill = $\text{Burden} \div 3$

- Rock Type = "HARD" THEN $\text{Actual Powder Factor} = (\text{Bench Height} - \text{Stemming Length} + \text{Subdrill}) \div (\text{Burden} \times \text{Spacing} \times \text{Bench Height})$

- IF RockType = "SOFT" THEN $\text{Actual Powder Factor} = (\text{Bench Height} - \text{Stemming Length} - \text{Subdrill}) \div (\text{Burden} \times \text{Spacing} \times \text{Bench Height})$

END IF OUTPUT Burden OUTPUT Stemming Length OUTPUT Actual

Powder Factor

2. Fragmentation Calculator

BEGIN Fragmentation Calculator

INPUT Next Step CONVERT Next Step TO UPPERCASE

IF Next Step = "NO" THEN OUTPUT "Thank You!"

ELSE IF Next Step = "YES"

THEN INPUT Bench Height

INPUT Relative Effective Energy Of Explosive INPUT Stemming Length 2

Mean Fragment Size = (Average In Hole Density × (Bench Height - Stemming Length 2)) ÷
Relative Effective Energy Of Explosive

OUTPUT Mean Fragment Size in "mm"

ELSE OUTPUT "Invalid!"

END IF

END Fragmentation Calculator